

Yunsoo Ha

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ACADEMIC POSITIONS

- **Cornell University** 2025 - Present
Postdoctoral Researcher in the School of Civil and Environmental Engineering USA
 - Research focus: simulation optimization for stochastic game-theoretic models
 - Research application: natural hazard insurance and municipal land use risk management
- **National Renewable Energy Laboratory** 2024 - 2025
Postdoctoral Researcher in the Computational Science Center USA
 - Research focus: multi-fidelity stochastic optimization, differentiable optimization
 - Research application: simulation-based adaptive traffic signal control

EDUCATION

- **Ph.D. in Industrial and Systems Engineering** 2018 - 2024
North Carolina State University USA
 - Advisor: Sara Shashaani
 - Committee: Yunan Liu, Reha Uzsoy, Quoc Tran-Dinh, Naihuan Jing
 - Dissertation: Expediting Stochastic Derivative Free Optimization
 - 🏆 ISE Distinguished Dissertation Award of the 2024 CA Anderson Awards
 - 🏆 Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award by IISE
- **Master of Operations Research** 2018 - 2021
North Carolina State University USA
- **M.S. in Logistics, School of Air Transport, Transportation, and Logistics** 2015 - 2017
Korea Aerospace University South Korea
- **B.S. in Logistics, School of Air Transport, Transportation, and Logistics** 2010 - 2015
Korea Aerospace University South Korea

RESEARCH INTEREST

Stochastic Optimization, Stochastic Simulation, Monte Carlo Methods, Decision-Focused Learning, Quantum Computing.

PUBLICATIONS

W=WORKING, C=CONFERENCE, J=JOURNAL

- [W.1] **Finding Multiple Equilibria in Nonconvex, Nonsmooth Multi-Region Insurance Markets under Hurricane Risk.**
Yunsoo Ha, Linda Nozick.
- [W.2] **Scalable Traffic Signal Control for Urban Mobility: A Subspace-Based Optimization Approach Using Deep Reinforcement Learning.**
Yunsoo Ha, Hyunwoo Shin, Julianne Mueller.
- [J.1] **Adaptive Regularization within Trust Region Methods for Stochastic Nonconvex Optimization.**
Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.
Under first-round review at *Mathematical Programming*
- [J.2] **Adaptive Sampling Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization.**
Yunsoo Ha, Julianne Mueller.
Under third-round review at *Mathematical Programming Computation* (submitted 2024).
- [J.3] **Complexity of Zeroth-and First-Order Stochastic Trust-Region Algorithms.**
Yunsoo Ha, Raghu Pasupathy, Sara Shashaani.
SIAM Journal on Optimization (2025).
- [J.4] **Two-Stage Estimation and Variance Modeling for Latency-Constrained Variational Quantum Algorithms.**
Yunsoo Ha, Sara Shashaani, Matt Menickelly.
INFORMS Journal on Computing (2025).

- [J.5] **Iteration Complexity and Finite-Time Efficiency of Adaptive Sampling Trust-Region Methods for Stochastic Derivative-Free Optimization.**
Yunsoo Ha, Sara Shashaani.
IIE Transactions (2025).
- [J.6] **Latency Considerations for Stochastic Optimizers in Variational Quantum Algorithms.**
Matt Menickelly, Yunsoo Ha, Matthew Otten.
Quantum (2023).
- [J.7] **A Decision Model to Determine the Number of Shuttles in a Tier-to-Tier SBS/RS.**
Yunsoo Ha, Junjae Chae.
International Journal of Production Research (2019).
- [J.8] **Free Balancing for a Shuttle-Based Storage and Retrieval System.**
Yunsoo Ha, Junjae Chae.
Simulation Modelling Practice and Theory (2018).
- [C.1] **Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling**
Yunsoo Ha, Juliane Mueller.
Winter Simulation Conference 2025.
- [C.2] **Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling.**
Yunsoo Ha, Sara Shashaani.
Winter Simulation Conference 2023.
- [C.3] **Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles With Adaptive Sampling.**
Yunsoo Ha, Sara Shashaani, Quoc Tran-Dinh.
Winter Simulation Conference 2021.

HONORS AND AWARDS

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|---|-----------------|
| • Second Place Winner of the 2025 Pritsker Doctoral Dissertation Award
<i>Institute of Industrial and Systems Engineers</i> | 2025 |
| • Outstanding Reviewer Award
<i>Winter Simulation Conference</i> 2024 | 2024 |
| • ISE Distinguished Dissertation Award of the 2024 CA Anderson Awards
<i>North Carolina State University</i> | 2024 |
| • Travel Awards for the 2023 Annual Midwest Optimization Meeting
<i>Michigan State University</i> | 2023 |
| • Mentored Teaching Fellowship
<i>North Carolina State University</i> | 2022 |
| • Scholarship for Excellent Academic Records
<i>Korea Aerospace University</i> | 2010, 2013-2016 |

RESEARCH EXPERIENCE

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|---|-----------------------|
| • Cornell University
<i>Postdoctoral Researcher (Mentors: Linda K. Nozick)</i> | 2025 – Present
USA |
| ◦ Developing simulation optimization methods for multi-region Cournot–Nash equilibrium problems in hurricane-insurance markets, advancing computational frameworks for stochastic and nonlinear equilibrium analysis. | |
| • National Renewable Energy Laboratory
<i>Postdoctoral Researcher (Mentors: Juliane Mueller and Devon Sigler)</i> | 2024 – 2025
USA |
| ◦ Developed a novel adaptive sampling stochastic trust region method for multi-fidelity stochastic optimization. | |
| ◦ Developed a second-order optimizer that uses diagonal Hessian approximations for deep learning applications. | |
| ◦ Developed a differentiable optimization algorithm for mixed-integer problems. | |
| ◦ Developed a subspace-based optimizer for large-scale traffic signal control by learning subspaces via deep RL. | |
| • North Carolina State University
<i>Research Assistant (Advisor: Sara Shashaani)</i> | 2019 – 2023
USA |
| ◦ Analyzed the computational complexities with and without Common Random Numbers (CRN) in stochastic optimization, and theoretically demonstrated that CRN can significantly reduce the computational burden. | |
| ◦ Enhanced the finite-time performance of the adaptive sampling trust-region method for simulation optimization through four key refinements: | |

- Improved the chances of identifying better solutions through the integration of direct search techniques,
- Constructed a quadratic model with diagonal Hessian within the trust region framework,
- Reused previously evaluated solutions and corresponding simulation outputs to reduce computational cost,
- Applied CRN to reduce the variance in function and gradient estimates.
- Developed simulation optimization solvers and problems from scratch and tested them using Python ( SimOpt).
- Developed a stochastic oracle for traffic signal control problems, analyzed its loss landscape characteristics, and evaluated the performance of various solvers in addressing the problem ( Poster).
- **Argonne National Laboratory** Summer 2022
Summer Intern (Mentors: Matt Menickelly and Jeffrey Larson) USA
 - Designed a gaussian process based trust region algorithm for noisy derivative-free optimization problems.
- **Argonne National Laboratory** Summer 2021
Summer Intern (Mentors: Matt Menickelly and Matt Otten) USA
 - Improved the randomized coordinate algorithm with adaptive sampling as a stochastic optimizer for variational hybrid quantum-classical algorithms.
- **Korea Aerospace University** 2015 - 2018
Researcher & Research Assistant (Advisor: Junjae Chae) South Korea
 - Developed original heuristics for machine operations, including shuttle's dynamic allocation.
 - Developed the decision model for the number of shuttles according to the demands.

TEACHING EXPERIENCE

- **ISE 362: Stochastic Models in Industrial Engineering** Spring 2023
Instructor North Carolina State University
- **ISE 441: Introduction to Simulation** Fall 2019, Spring 2020, Fall 2020, and Fall 2022
Teaching Assistant North Carolina State University
- **ISE 748: Quality Engineering** Spring 2019
Teaching Assistant North Carolina State University
- **ISE 498: Senior Design Project** Fall 2018
Teaching Assistant North Carolina State University
- **Analysis of Logistics System** Fall 2015 and Fall 2016
Teaching Assistant Korea Aerospace University

MENTORING EXPERIENCE

- **Graduate Students**
 - Qian Ni (Ph.D. Student at Cornell University CEE) Fall 2025 – Present
 - Hyunwoo Shin (Ph.D. Candidate at Virginia Tech ISE) Summer 2025 – Present
- **Undergraduate Students**
 - Kevin Xu (NCSU Statistics) Fall 2022
 - Wes Hankinson (NCSU ISE) Spring 2022

PRESENTATION

Invited Talks

- **INFORMS Optimization Society Conference 2026** 2026
A Derivative-Free Solver for Multi-Region Stochastic Equilibrium Problems
- **2025 INFORMS Virginia Tech Student Chapter Seminar Series** 2025
Expediting Simulation Optimization
- **2025 INFORMS Annual Meeting** 2025
Adaptive Sampling-Based Bi-Fidelity Stochastic Trust Region Method for Derivative-Free Stochastic Optimization
- **International Symposium on Mathematical Programming** 2024
Two talks: Multi-Fidelity Stochastic Trust-Region Method, Is Building First-Order Simulation Oracles Really Worth It?
- **INFORMS Optimization Society Conference 2024** 2024
First-Order Trust-Region Methods with Adaptive Sampling.
- **2023 Annual Midwest Optimization Meeting** 2023
Common Random Numbers and Complexity in Simulation Optimization with Adaptive Sampling.

- **2023 INFORMS Annual Meeting** 2023
Consistency and Complexity of Adaptive Sampling Based Trust-Region Optimization.
- **Modeling and Optimization: Theory and Applications (MOPTA) 2023** 2023
Expediting Stochastic Derivative Free Optimization.
- **2022 INFORMS Annual Meeting** 2022
Complexity Analysis of Trust-Region Optimization with Adaptive Sampling for Zeroth-Order Stochastic Oracles.
- **INFORMS Optimization Society Conference 2022** 2022
Adaptive Sampling Trust-Region Optimization with Diagonal Hessian for Derivative-Free Stochastic Oracles.

Contributed Talks

- **Winter Simulation Conference 2025** 2025
Multi-Fidelity Stochastic Trust Region Method with Adaptive Sampling.
- **Institute of Industrial and Systems Engineers Annual Conference 2025** 2025
First-Order Trust-Region Methods with Adaptive Sampling.
- **Winter Simulation Conference 2023** 2023
Towards Greener Stochastic Derivative-Free Optimization with Trust Regions and Adaptive Sampling.
- **Winter Simulation Conference 2021** 2021
Improved Complexity of Trust-Region Optimization for Zeroth-Order Stochastic Oracles with Adaptive Sampling.
- **Winter Simulation Conference 2020** 2020
Traffic Signal Control Simulation and Optimization (Poster).

ACADEMIC SERVICE

- **Journal Refereeing**
 - Journal of Simulation
 - Journal of Scientific Computing
 - Computational Optimization and Applications
- **Conference Refereeing**
 - Winter Simulation Conference 2024-2026
- **Session Chair**
 - 2022, 2023, 2025 INFORMS Annual Meeting
 - Modeling and Optimization: Theory and Applications (MOPTA) 2023
 - Winter Simulation Conference 2023